Hrishikesh S Raj

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Summary

Software Engineer bridging the gap between Geospatial AI and robust application architecture. Experienced in deploying transformer-based vision models, optimizing complex data pipelines, and building end-to-end full-stack systems with offline-capable logic.

Technical Skills

Programming Languages: Java, Python, C++, SQL

Frameworks & Libraries: PyTorch, Scikit-learn, Pandas, React Native

Tools & Platforms: Git, Visual Studio Code, Vercel, Render

Domain Knowledge: Data Structures & Algorithms (DSA), Design & Analysis of Algorithms (DAA), Machine Learning, Database Design, API Design

Projects

Satellite Image Segmentation Using Transformers| *GitHub* — Jan 2026 – May 2026
Deep Learning / Remote Sensing — *Python, PyTorch, SegFormer, NumPy, Rasterio*

- Core Architecture: Developed an uncertainty-aware semantic segmentation pipeline for high-resolution satellite imagery using transformer-based architectures, specifically engineering an Evidential Deep Learning (EDL) SegFormer model.
- Preprocessed multi-spectral geospatial raster data using Rasterio; implemented patch-based training to handle large-scale satellite tiles efficiently
- Quantifiable Results: Achieved a verified 65.81% mIoU on the DeepGlobe Land Cover dataset, establishing superior failure localization and uncertainty estimation accuracy compared to traditional Monte Carlo (MC) Dropout models.
- Advanced Optimization: Formulated a custom loss function incorporating KL divergence annealing and designed an Out-of-Distribution (OOD) detection pipeline to validate model robustness against synthetic corruptions.
- Academic Output: Documented experimental methodologies and findings into a technical research paper successfully submitted to the ICIET2026 conference.

Landslide Prediction with ML | *GitHub* — Aug 2025 – Dec 2025
Geospatial AI & Disaster Management — *Python, Scikit-learn, NumPy, Rasterio, CesiumJS*

- Built a browser-based 3D geospatial demo predicting landslide-prone areas in Mundakkai, Wayanad using ML on terrain, satellite imagery, and environmental data
- Trained in K-means to generate susceptibility heatmaps overlaid on CesiumJS terrain for interactive disaster risk exploration
- Completed as a proof-of-concept for real-time environmental simulation and risk communication

Education

SRM Institute of Science and Technology – Kattankulathur — Chennai, India
B.Tech, Computer Science and Engineering | CGPA: 7.35/10 — *Aug 2022 – May 2026*

Dayapuram Residential School — Calicut, Kerala
Class XII – CBSE (CMPC) | Percentage: 83.2% — *Graduated 2021*

Languages

English *(Professional Working Proficiency)*, Malayalam *(Native)*, Hindi *(Limited Working Proficiency)*